

University of Asia Pacific

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FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text

A privacy-preserving federated learning architecture for cross-institutional Bangla clinical text mining

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INTRODUCTION

Bangla Medical Text

Bangla Clinical Text is Exploding, but Underrepresented in Medical NLP.

Privacy Challenge

Patient Data Cannot Be Centralized Across Hospitals.

FedMed-BN

Federated Learning as a Privacy-Preserving Way Forward.

MOTIVATION



Language Gap

No medical AI system exists for Bangla clinical text.



Data Privacy

Data cannot be centralized



Security Risk

Gradient leakage risk

LITERATURE REVIEW

Year	Paper	Dataset	Model	Performance	Limitations
2019	Hasan et al. – Medical Prescription NLP via OCR [2]	~800K prescription text rows	OCR (EAST+Tesseract) + Rule-based NLP	93–95% accuracy	Manual rule creation; prescriptions only; not federated
2019	Dligach & Miller – Two-Stage Federated Phenotyping [3]	MIMIC-III; i2b2 Obesity Challenge	DAN + Federated Phenotyping Classifier	F1 = 0.724 ($\approx$ centralized)	No differential privacy; weak on rare conditions
2022	Islam et al. – BanglaBioMed NER [5]	BanglaBioMed (818 sent., 11,196 tokens)	BanglaBERT / mBERT / XLM-RoBERTa	BanglaBERT F1 = 81.33% (best)	No federated learning; no privacy; small dataset
2022	Liu et al. – FedInitPrune [7]	Fudan Chinese Text; THUCNews	Federated TextCNN + Parameter Pruning	74.26% comm. reduction, <2% accuracy drop	No privacy mechanism; Chinese text only
2023	Huang et al. – Federated ICD-10 Classification [6]	3 Taiwanese hospital EMR silos	Federated PubMedBERT / BioBERT / ClinicalBERT	F1 = 0.812 (+3.11%)	No DP; high local compute; non-IID challenges
2023	Lyu et al. – LDP for Edge NLP [10]	IMDB Reviews; MovieLens	LDP + Bitwise Quantization + Feature Hashing	Competitive accuracy, low resource cost	LDP noise reduces accuracy; not tested on medical text

LITERATURE REVIEW (Cont...)

Year	Paper	Dataset	Model	Performance	Limitations
2024	Chen et al. – FLCMC [1]	IMCS-V2 (4,116 dialogues)	FedPA / FedPAP	FedPAP: 0.79 accuracy	No DP; Chinese only; large model impractical on edge
2024	Rui Wang, Ricardo Henao – Gradient-Free Prompt Tuning [4]	SST-2, AG's News, TREC	RoBERTa-large + Genetic Algorithm	1000× lower comm. cost, competitive accuracy	GA computationally heavy; prompt-template dependent
2024	Xun Li – FL-NLP Systematic Review [9]	Gboard, Stack Overflow, IMDB, etc.	Survey: FedAvg, FedProx, FedAdam, BERT	92% spam; 76% sentiment classification	Survey only; non-English NLP underexplored
2024	Ju Sun, and Rui Zhang – FL Biomedical NLP Evaluation [14]	8 biomedical NER / RE corpora	FedAvg / FedProx + BERT variants	FedProx F1 = 0.994 (non-IID)	Simulated splits; no DP or homomorphic encryption
2025	Ahmed Al-Ali – FL for Healthcare Big Data [8]	TB Chest X-ray (7,000 images)	ResNet + FedAvg + SMOTE	96.7% accuracy, F1 = 97.4%	Imaging only; no DP; no NLP component
2025	Nilesh Patel and Deep Shah – Edge-Aggregated FL Healthcare [11]	Synthetic data + MIMIC-III ICU	Hierarchical FL + Shamir's Secret Sharing	Strong privacy via SMPC	Tabular only; no NLP; synchronous rounds

LITERATURE REVIEW (Cont...)

Year	Paper	Dataset	Model	Performance	Limitations
2025	Johan van Soest, Michel Dumontier – Personal Health Train [12]	12 hospitals, 8 countries (CT scans)	FedAvg + PHT / Vantage6	Feasible across 12 real hospitals	Imaging only; network bottlenecks; no formal DP
2025	K. Srinivasa Rao and P. S. N. Murthy – FedHealth-NLP [13]	4 community health text corpora	FedBERT + DP + Knowledge Graph NN	91.3% accuracy; 42% comm. reduction	Simulated nodes; DP cuts accuracy by 0.8%; English only
2025	Jian Gao, Chen Liang, and Fei Wang – FedCliMask [15]	12,996 synthetic clinical notes	UMLS-guided Semantic Masking + ClinicalBERT	Optimal privacy-utility balance identified	Not implemented; synthetic & English only

EVALUATION OF RELATED WORK

2019

OCR + rule-based NLP for prescriptions; first federated phenotyping (MIMIC-III)

2022

BanglaBioMed NER baseline; communication-efficient FL (FedInitPrune)

2023

Federated ICD-10 classification on real hospital silos; LDP for edge NLP

2024

Chinese federated dialogue (FLCMC); FL-NLP surveys; FL beats few-shot LLMs on biomedical NLP

2025

FL scales to imaging + text health data; DP-integrated FedHealth-NLP; ontology-aware masking (FedCliMask)

RESEARCH GAP & KEY CONTRIBUTION

Research Gap

- No federated learning for Bangla medical text
- Privacy protection is still limited
- Bangla medical NLP uses centralized data
- No solution for low-resource Bangla healthcare

Key Contribution

- First federated Bangla medical NLP
- Privacy preserving learning
- Realistic hospital simulation
- Reusable benchmark for future research

EXPERT FEEDBACK

Requirement validation interviews were held with three clinical experts:

Expert	Occupation	Key Feedback
Shayela Begum	Physician	Recommended incorporating real-world prescription phrasing to improve clinical relevance.
Zakir Hossain	Surgeon	Emphasized clearly documenting what data leaves the client vs. what stays local.
Nasima Kamal	Doctor	Recommended prioritizing evaluation on realistic non-IID clinic data distributions.

Overall consensus: federated + privacy-preserving design is clinically relevant and directly useful for Bangladeshi hospitals.

DATASET DESCRIPTION

Dataset	Source / Year	Contents	Role in FedMed-BN
Bangla MedER	PLOS One, Sept 2025	6,895 real-world medical observations with disease / specialty labels	Primary dataset, partitioned into 5 non-IID hospital clients
BanglaBioMed	ACL BioNLP, 2022	818 sentences, 11,196 tokens; 4 biomedical entity types	Supports biomedical NER and improves domain knowledge
Bangla-MedER (Mendeley)	Mendeley Data, Oct 2025	2,980 annotated statements, 14,357 tokens; 6 entity types	Provides additional annotated medical entities

6,895

medical records

5

hospital clients

6

medical entity types

80/20

train-test split per client

DATASET SAMPLES

BanglaBioMed		
Structure of Entity	Entity	Sentence
Simple Entity (single and multi-word)	জ্বর, কাশি, গলাব্যথা, শ্বাসকষ্ট	জ্বর, কাশি, গলাব্যথা, শ্বাসকষ্ট হচ্ছে অমিক্রনের মূল উপসর্গ।
	Fever, cough, sore throat, shortness of breath (<i>English Translation</i>)	Fever, cough, sore throat, shortness of breath are the main symptoms of Omicron. (<i>English Translation</i>)
Complex Entity (Overlapping)	নাক দিয়ে রক্ত আসা, নাক দিয়ে পানি আসা	নাক দিয়ে রক্ত বা পানি আসা
	Blood coming through the nose, Water coming through the nose (<i>English Translation</i>)	Blood or water coming through the nose (<i>English Translation</i>)

Figure 1: Examples of entities representing varied structures

Bangla MedER							
1	Medical Text	Medicine\ Chemical N	Organ	Disease	Hormone	Pharmacological Class	Common Medical Terms
2	ফেনোক্সোফেনাডিন চুলকানি, হাঁচি এবং জ্বালা সহ অ্যালার্জির লক্ষণগুলির চিকিত্সা করে। এই লক্ষণগুলির	ফেনোক্সোফেনাডিন	চোখ, নাক	চুলকানি, হাঁচি, জ্বালা, চুলকানি,	হিস্টামিন		
3	টাফনিল হচ্ছে মূলত মাইগ্রেশন এর মত ব্যথা কে উপশম করতে ব্যবহার করা হয়। এছাড়াও মাইগ্রেনের	টাফনিল		ব্যথা, জ্বর			
4	অ্যামোক্সিসিলিন হল একটি পেনিসিলিন-ভিত্তিক অ্যান্টিবায়োটিক (অ্যামিনো-পেনিসিলিন) যা ব্যাকটেরিয়	অ্যামোক্সিসিলিন		ব্যাকটেরিয়া সংক্রমণ		অ্যামিনো-পেনিসিলিন	অ্যান্টিবায়োটিক
5	অ্যাস্ট্রোপি ন হল একটি ট্রিপেন অ্যালকালয়েড যা সাধারণত কমাতে ব্যবহৃত হয় ব্যথা এবং প্রদাহ এটি শরী	অ্যাস্ট্রোপি ন		প্রদাহ, জ্বর, ব্যথা		ট্রিপেন অ্যালকালয়েড	
6	অ্যাজিথ্রোমাইসিন ওষুধটি ম্যাক্রোলাইড অ্যান্টিবায়োটিক নামক ওষুধের একটি গ্রুপের অন্তর্গত, যা সংক্র	অ্যাজিথ্রোমাইসিন	দ্বক, শ্বাসযন্ত্র	নিউমোনিয়া		ম্যাক্রোলাইড অ্যান্টিবায়োটিক	অ্যান্টিবায়োটিক, সংক্রমণ
7	অ্যালবেনডাজল, একটি শক্তিশালী অ্যান্টিপ্যারাসাইটিক ওষুধ, বিভিন্ন চিকিৎসায় এর কার্যকারিতার জন্য	অ্যালবেনডাজল		কৃমি উপদ্রব		অ্যান্টিপ্যারাসাইটিক	ওষুধ
8	ক্লোরফেনিরামাইন হল একটি অ্যান্টিহিস্টামাইন ওষুধ যা মৌসুমি অ্যালার্জি, অ্যালার্জিক রাইনাইটিস, আ	ক্লোরফেনিরামাইন	নাক, চোখ	অ্যালার্জি, হাঁচি		অ্যান্টিহিস্টামাইন	
9	ডাইক্লোফেনাক একটি নন-স্টেরয়েডাল অ্যান্টি-ইনফ্ল্যামেটরি ড্রাগ (NSAID)। এটি মূল কারণ বা শরীরে ব	ডাইক্লোফেনাক	দ্বক	প্রদাহ, ব্যথা		অ্যান্টি-ইনফ্ল্যামেটরি	ব্যথানাশক
10	ডাইক্লোফেনাক বিভিন্ন অ্যালার্জির প্রতিক্রিয়ার লক্ষণ দেখাতে পারে, যেমন শ্বাস নিতে অসুবিধা, মুখ এবং	ডাইক্লোফেনাক	মুখ, গলা, দ্বক	দ্বকের ব্যথা, ফোসকা			ব্যথানাশক
11	ডাইক্লোফেনাক সমস্ত ওষুধগুলি কক্ষ তাপমাত্রায় সংরক্ষণ করা উচিত এবং সরাসরি তাপ এবং আলো থে	ডাইক্লোফেনাক					ঔষধ
12	এসোমেপ্রাজল, একটি শক্তিশালী ওষুধ, বৈপ্লবিক পরিবর্তন করেছে গ্যাস্ট্রিক সমস্যার চিকিত্সা। এই অসা	এসোমেপ্রাজল		গ্যাস্ট্রিক, আলসার			
13	প্রতিদিন একই সময়ে এসোমেপ্রাজল নিন। প্রেসক্রিপশন বা প্যাকেজ নির্দেশাবলী সাবধানে অনুসরণ করুন। নির্ধারিত ডোজ বা সময়কাল অতিক্রম	এসোমেপ্রাজল		গ্যাস্ট্রিক, আলসার			
14	অ্যাসিক্লোফেন্যাক অ স্টেরয়েডাল অ্যান্টি-ইনফ্ল্যামেটরি ড্রাগস (NSAIDs) বিভাগের অন্তর্গত। দীর্ঘস্থায়ী	অ্যাসিক্লোফেন্যাক		দীর্ঘস্থায়ী প্রদাহ, ব্যথা	প্রোস্টাগ্ল্যান্ডিন	অ স্টেরয়েডাল অ্যান্টি-ইনফ্ল্যামেটরি ড্রাগস	
15	শরীরে যে কোনো ধরনের ব্যথা অনুভব করলে অবশ্যই প্যারাসিটামল খেতে হবে। আপনার জ্বর হলে তা	প্যারাসিটামল		ব্যথা, জ্বর			ঔষধ, ডাক্তার
16	প্যারাসিটামল একটি ব্যথা উপশমকারী এবং সেইসাথে একটি অ্যান্টিপাইরেটিক (যা শরীরের তাপমাত্রা ক	প্যারাসিটামল		ব্যথা	প্রোস্টাগ্ল্যান্ডিনস	অ্যান্টিপাইরেটিক	ওষুধ
17	টাফনিল প্রধানত মাইগ্রেশনের মতো ব্যথা উপশম করতে ব্যবহৃত হয়। মাইগ্রেন সহ জ্বর সহ বিভিন্ন ধর	টাফনিল		ব্যথা, জ্বর			
18	ক্যালসিট্রিওল হল ভিটামিন ডি-এর একটি জৈবিকভাবে সক্রিয় কৃত্রিম রূপ। এটি দীর্ঘস্থায়ী কিডনি ব্যর্থতা	ক্যালসিট্রিওল	কিডনি	হাড়ের রোগ			
19	ভিটামিন বি কমপ্লেক্স আমাদের শরীরের স্বাস্থ্য বজায় রাখতে গুরুত্বপূর্ণ ভূমিকা পালন করে, তবে যেহেতু	ভিটামিন বি কমপ্লেক্স				ভিটামিন	ভিটামিন বি কমপ্লেক্স, স্বাস্থ্য, অভাব, খাদ্য

Annotated

Bangla Medical Sentence

রোগীর ডায়াবেটিস আছে এবং ইনসুলিন ব্যবহার করছে।

↓

Annotated

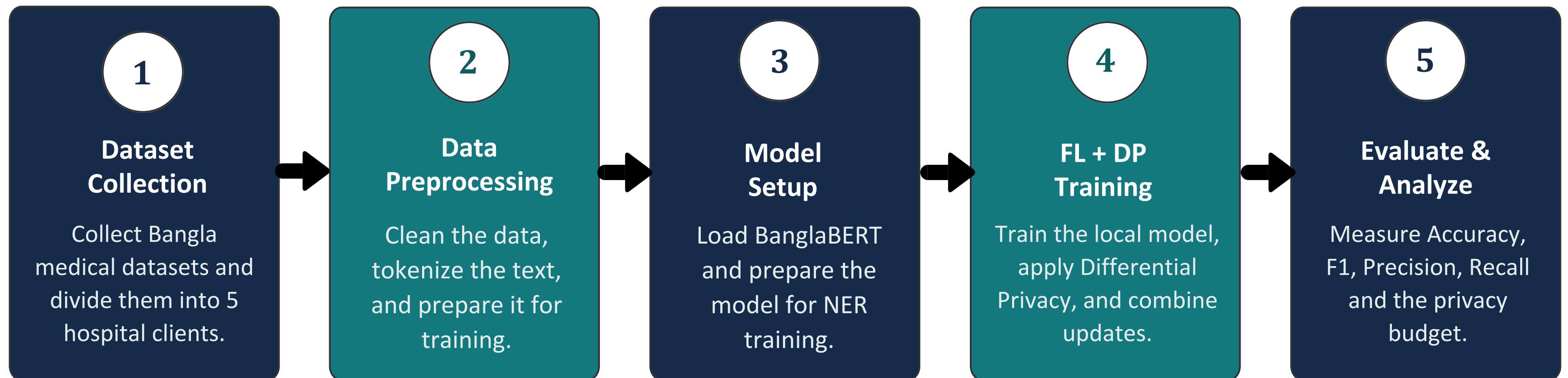
ডায়াবেটিস

ইনসুলিন

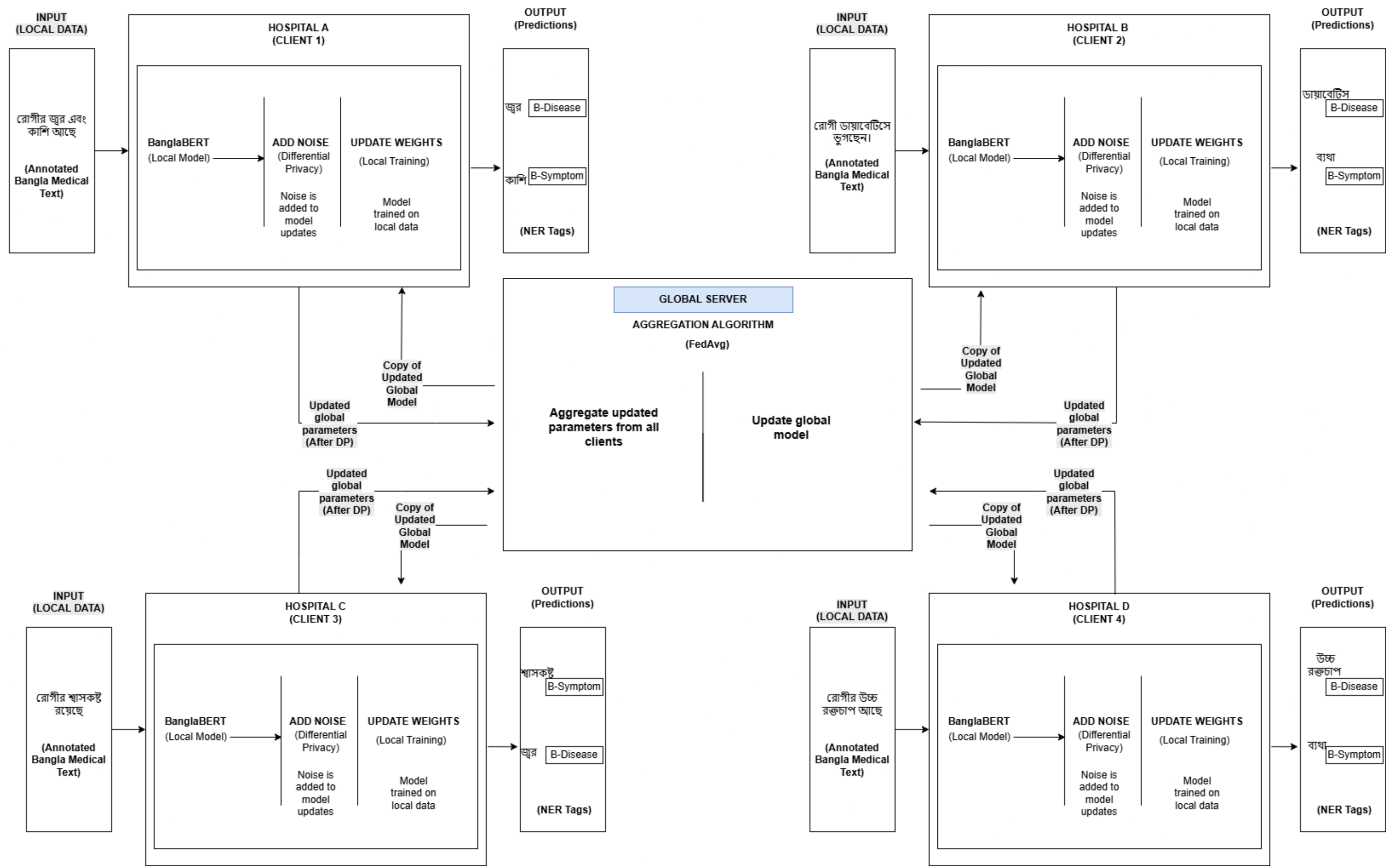
Disease

Medicine

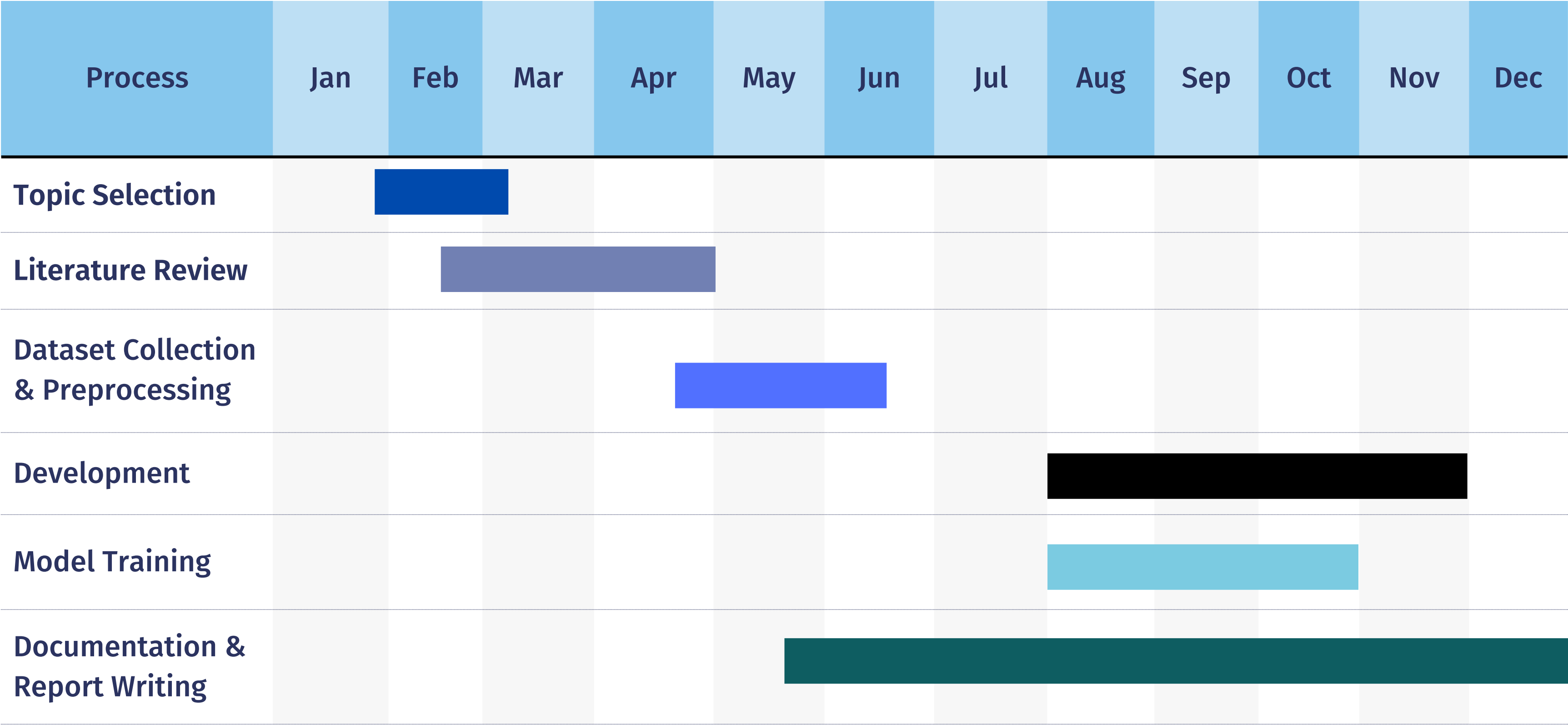
METHODOLOGY



METHODOLOGY FRAMEWORK



TIMELINE





CONCLUSION

- **Introduced FedMed-BN for Bangla Medical NER.**
- **Found a key gap in Bangla Medical NLP privacy.**
- **Designed architecture combining BanglaBERT with DP-FedAvg.**
- **Enabled privacy-preserving Bangla healthcare NLP.**





FUTURE WORK

- **Implement DP-FedAvg pipeline.**
- **Run experiments across 5 simulated hospital clients.**
- **Evaluate accuracy, F1, precision, recall.**
- **Expand dataset with diverse Bangla medical sources**
- **Explore secure aggregation and personalized FL.**



CEP MAPPING



How Knowledge Profiles (Ks) are addressed through FedMed-BN, and mapping among Ks, CLOs, and PLOs

Ks	Attribute	How Ks is Addressed Through This Research	Evidence	CLOs	PLOs
K3	Engineering Fundamentals	Applies Python programming, distributed computing principles, and client-server communication as the engineering foundation of the federated learning framework.	Ch.3 – System Architecture; Ch.4.2 – Tools, Libraries & Technologies	CLO2	PLO(b)
		Applies machine learning fundamentals, including transformer-based NLP and BanglaBERT, for Bangla medical text processing.	Ch.2.1 – Preliminaries; Ch.3 – Methodology Framework	CLO2	PLO(c)
K4	Specialist Knowledge	Applies Federated Learning, Differential Privacy, BanglaBERT, and Medical Named Entity Recognition to design a privacy-preserving Bangla medical NLP framework.	Ch.2.1 – Preliminaries; Ch.3.2.1 – High-Level Architecture; Ch.3.2.3 – Design Alternatives	CLO2	PLO(b)

CEP MAPPING (Cont...)



How Knowledge Profiles (Ks) are addressed through FedMed-BN, and mapping among Ks, CLOs, and PLOs

Ks	Attribute	How Ks is Addressed Through This Research	Evidence	CLOs	PLOs
K5	Engineering Design	Designs the FedMed-BN architecture by integrating local client training, Differential Privacy, and FedAvg aggregation.	Ch.3.2.1 – High-Level Architecture	CLO2	PLO(b)
		Designs secure communication between hospital clients and the global server while considering Non-IID data distribution and parameter aggregation.	Ch.3.2.2 – Detailed Design Diagrams	CLO2	PLO(c)
K6	Engineering Practice	Plans implementation using PyTorch, Flower Framework, Hugging Face Transformers, and Opacus following modern engineering practices.	Ch.4.2 – Tools & Technologies; Ch.4.5 – Model Development	CLO2	PLO(c)

CEP MAPPING (Cont...)



How Complex Engineering Problem attributes (Ps) are addressed through FedMed-BN, and mapping among Ps, CLOs, and PLOs

Ps	Attribute	How Ps is Addressed Through This Research	Evidence	CLOs	PLOs
P1	Depth of Knowledge Required	Integrates knowledge of Federated Learning, Differential Privacy, BanglaBERT, and Bangla medical NLP to formulate a privacy-preserving distributed learning framework.	Ch.2.1–2.4 – Background, Related Works & Research Gap	CLO1	PLO(I)
		Applies engineering fundamentals to design the complete federated learning pipeline for Bangla medical text analysis.	Ch.3.2.1–3.2.3 – Architecture & Design Alternatives	CLO2	PLO(c)

CEP MAPPING (Cont...)



How Complex Engineering Problem attributes (Ps) are addressed through FedMed-BN, and mapping among Ps, CLOs, and PLOs

Ps	Attribute	How Ps is Addressed Through This Research	Evidence	CLOs	PLOs
P3	Depth of Analysis Required	Analyses different Federated Learning architectures, Differential Privacy mechanisms, and Bangla language models before selecting the proposed approach.	Ch.2.2–2.4 – Literature Review, Comparative Analysis & Research Gap	CLO2	PLO(b)
		Evaluates the privacy–utility trade-off and the impact of Non-IID client distributions while designing the proposed framework.	Ch.3.2.3 – Design Alternatives; Ch.3.3 – Algorithmic Flow	CLO2	PLO(c)
P6	Extent of Stakeholder Involvement	Identifies stakeholder requirements from healthcare organizations and considers privacy, usability, and regulatory needs during system requirement analysis.	Ch.3.1 – Requirement Analysis & Stakeholder Questionnaire	CLO2	PLO(b)

CEA MAPPING

How Complex Engineering Activity attributes (As) are addressed through FedMed-BN, and mapping among As, CLOs, and PLOs



As	Attribute	How As is Addressed Through This Research	Evidence	CLOs	PLOs
A1	Range of Resources	Uses Bangla medical datasets, research publications, GPU computing, Federated Learning frameworks, NLP libraries, and healthcare domain knowledge to develop the proposed system.	Ch.2.2–2.3 – Literature Review; Ch.4.2 – Tools & Technologies	CLO1	PLO(l)
		Selects software tools, hardware resources, and project planning techniques to support implementation of the proposed system.	Ch.4.7 – Hardware & Software Requirements; Ch.6 – Timeline & Budget	CLO3	PLO(k)
A3	Innovation	Combines BanglaBERT, Federated Learning, and Differential Privacy into a unified framework for Bangla medical NLP, addressing the identified research gap.	Ch.3.2.3 – Design Alternatives & Rationale; Ch.1.6 – Expected Outcomes	CLO2	PLO(c)

CEA MAPPING (Cont...)

How Complex Engineering Activity attributes (As) are addressed through FedMed-BN, and mapping among As, CLOs, and PLOs



As	Attribute	How As is Addressed Through This Research	Evidence	CLOs	PLOs
A4	Consequences for Society and Environment	Assesses the ethical, legal, and societal impact of privacy-preserving healthcare AI by protecting sensitive patient information.	Ch.1.7.1–1.7.2 – Societal, Health, Safety & Legal Impact	CLO5	PLO(f)
		Considers sustainability and environmental issues related to federated model training and computational resource usage.	Ch.1.7.3 – Environmental & Sustainability Impact	CLO4	PLO(g)

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THANK YOU



Q & A Session



EXPERT CONSULTATION EVIDENCE

Responses cannot be edited

FedMed-BN: Expert Evaluation Questionnaire (NLP & Medical Reviewers)

FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text
This thesis develops a Federated Learning framework for Banglish (mixed Bangla-English) next-word prediction using BanglishBERT, enabling multiple clients to collaboratively train a medical-text language model without sharing raw, sensitive patient data with a central server.

Thesis Students: Faiza Haque Shoily, Muhib Al Ahad, Tanjuma Noor
Supervisor: Shammi Akhtar
Co-Supervisor: Md Shafayatul Haque

Scoring Scale: 0 = No valid | 1 = Incorrect | 2 = Partially correct | 3 = Mostly correct | 4 = Completely correct

Please rate each statement below on the 0–4 scale.

The respondent's email (zplastic61@gmail.com) was recorded on submission of this form.

* Indicates required question

Name of the Evaluator *

Zakir Hossain

Occupation *

Physician

Contact Info (GitHub link, E-mail, LinkedIn, etc.) *

zplastic61@gmail.com

Questionnaire Introduction and Evaluator Details (Zakir Hossain)

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? *

01234

0 - No valid

4 - Completely correct

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? *

01234

0 - No valid

4 - Completely correct

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? *

01234

0 - No valid

4 - Completely correct

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? *

01234

0 - No valid

4 - Completely correct

Section 1: Correct Application of Subject-Specific Concepts

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? *

01234

0 - No valid

4 - Completely correct

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? *

01234

0 - No valid

4 - Completely correct

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? *

01234

0 - No valid

4 - Completely correct

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? *

01234

0 - No valid

4 - Completely correct

Section 2: Logical Consistency

EXPERT CONSULTATION EVIDENCE (Cont...)

3. Validity and Justification

These questions probe whether design and evaluation decisions are adequately justified with evidence or precedent from the literature.

Is there sufficient justification for choosing BanglishBERT over alternatives such as mBERT or XLM-RoBERTa? *

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0 - No valid

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4 - Completely correct

Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used? *

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0 - No valid

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4 - Completely correct

Is the validity of the simulated client partitioning method adequately explained and justified? *

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0 - No valid

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4 - Completely correct

Are performance metrics (accuracy, perplexity) appropriate and justified for evaluating next-word prediction specifically? *

0

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0 - No valid

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4 - Completely correct

4. Innovation and Contribution

These questions assess the originality and practical value of FedMed-BN relative to existing federated NLP and Bangla medical NLP research.

Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks? *

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0 - No valid

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4 - Completely correct

Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource/non-English languages)? *

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0 - No valid

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4 - Completely correct

Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context? *

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0 - No valid

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4 - Completely correct

5. Overall Quality

These questions ask for a holistic judgment of technical soundness, coherence, and contribution of the thesis as a whole.

Overall, does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem? *

0

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0 - No valid

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4 - Completely correct

Are the project's stated limitations and future work realistic and proportionate to its current stage? *

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0 - No valid

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4 - Completely correct

Would this work be considered a meaningful and publishable contribution to federated NLP for low-resource medical text? *

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0 - Strongly Disagree

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4 - Strongly Agree

Comments or Additional Suggestions *

Section 3: Validity and Justification

Section 4: Innovation and Contribution

Section 5: Overall Quality

EXPERT CONSULTATION EVIDENCE (Cont...)

Responses cannot be edited

FedMed-BN: Expert Evaluation Questionnaire (NLP & Medical Reviewers)

FedMed-BN: Privacy-Preserving Federated Learning for Bangla Medical Text
This thesis develops a Federated Learning framework for Banglish (mixed Bangla-English) next-word prediction using BanglishBERT, enabling multiple clients to collaboratively train a medical-text language model without sharing raw, sensitive patient data with a central server.

Thesis Students: Faiza Haque Shoily, Muhib Al Ahad, Tanjuma Noor
Supervisor: Shammi Akhtar
Co-Supervisor: Md Shafayatul Haque

Scoring Scale: 0 = No valid | 1 = Incorrect | 2 = Partially correct | 3 = Mostly correct | 4 = Completely correct

Please rate each statement below on the 0–4 scale.

The respondent's email (shayelabegum69@gmail.com) was recorded on submission of this form.

* Indicates required question

Name of the Evaluator *

Shayela Begum

Occupation *

Physician

Contact Info (GitHub link, E-mail, LinkedIn, etc.) *

shayelabegum69@gmail.com

Questionnaire Introduction and Evaluator Details (Shayela Begum)

1. Correct Application of Subject-Specific Concepts

These questions assess whether NLP and Federated Learning concepts, and Bangla/Banglish medical-domain knowledge, are applied correctly in FedMed-BN.

Does the project correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than simply simulating distributed training without true data isolation? *

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0 - No valid

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4 - Completely correct

Is BanglishBERT an appropriate and correctly used pretrained model for code-mixed Bangla-English medical text? *

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0 - No valid

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4 - Completely correct

Is the FedAvg aggregation mechanism implemented and explained correctly (e.g., weighted averaging by client data size)? *

0

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0 - No valid

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4 - Completely correct

Does the project correctly distinguish next-word prediction (language modelling) from related but different tasks like text classification or NER? *

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0 - No valid

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4 - Completely correct

Section 1: Correct Application of Subject-Specific Concepts

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? *

0

1

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3

4

0 - No valid

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4 - Completely correct

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? *

0

1

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4

0 - No valid

☐

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☐

4 - Completely correct

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? *

0

1

2

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4

0 - No valid

☐

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☐

4 - Completely correct

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? *

0

1

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4

0 - No valid

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4 - Completely correct

Section 2: Logical Consistency

EXPERT CONSULTATION EVIDENCE (Cont...)

3. Validity and Justification

These questions probe whether design and evaluation decisions are adequately justified with evidence or precedent from the literature.

Is there sufficient justification for choosing BanglishBERT over alternatives such as mBERT or XLM-RoBERTa? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Is the validity of the simulated client partitioning method adequately explained and justified? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Are performance metrics (accuracy, perplexity) appropriate and justified for evaluating next-word prediction specifically? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

4. Innovation and Contribution

These questions assess the originality and practical value of FedMed-BN relative to existing federated NLP and Bangla medical NLP research.

Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource/non-English languages)? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

5. Overall Quality

These questions ask for a holistic judgment of technical soundness, coherence, and contribution of the thesis as a whole.

Overall, does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Are the project's stated limitations and future work realistic and proportionate to its current stage? *

0 1 2 3 4
0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Would this work be considered a meaningful and publishable contribution to federated NLP for low-resource medical text? *

0 1 2 3 4
0 - Strongly Disagree ☐ ☐ ☐ ☒ ☐ 4 - Strongly Agree

Comments or Additional Suggestions *

.....

Section 3: Validity and Justification

Section 4: Innovation and Contribution

Section 5: Overall Quality

EXPERT CONSULTATION EVIDENCE (Cont...)

Responses cannot be edited

FedMed-BN: Expert Evaluation Questionnaire (NLP & Medical Reviewers)

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Thesis Students: Faiza Haque Shoily, Muhib Al Ahad, Tanjuma Noor
Supervisor: Shammi Akhtar
Co-Supervisor: Md Shafayatul Haque

Scoring Scale: 0 = No valid | 1 = Incorrect | 2 = Partially correct | 3 = Mostly correct | 4 = Completely correct

Please rate each statement below on the 0–4 scale.

The respondent's email (nasimakamal1972@gmail.com) was recorded on submission of this form.

** Indicates required question*

Name of the Evaluator ***
Nasima Kamal

Occupation ***
Doctor

Contact Info (GitHub link, E-mail, LinkedIn, etc.) ***
nasimakamal1972@gmail.com

Questionnaire Introduction and Evaluator Details
(Nasima Kamal)

1. Correct Application of Subject-Specific Concepts

These questions assess whether NLP and Federated Learning concepts, and Bangla/Banglish medical-domain knowledge, are applied correctly in FedMed-BN.

Does the project correctly apply Federated Learning principles (local training, parameter sharing, central aggregation) rather than simply simulating distributed training without true data isolation? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☒ ☐ ☐ 4 - Completely correct

Is BanglishBERT an appropriate and correctly used pretrained model for code-mixed Bangla-English medical text? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☐ ☐ ☒ 4 - Completely correct

Is the FedAvg aggregation mechanism implemented and explained correctly (e.g., weighted averaging by client data size)? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☒ ☐ ☐ 4 - Completely correct

Does the project correctly distinguish next-word prediction (language modelling) from related but different tasks like text classification or NER? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☒ ☐ ☐ 4 - Completely correct

Section 1: Correct Application of Subject-Specific Concepts

2. Logical Consistency

These questions check whether the framework's design choices follow logically from the stated problem and from each other.

Does the choice of a federated (rather than centralized) architecture logically follow from the privacy-sensitivity of medical text? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☐ ☒ ☐ 4 - Completely correct

Is the use of a non-IID (heterogeneous) client data simulation logically consistent with how real Bangladeshi clinics/devices would differ from each other? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☒ ☐ ☐ 4 - Completely correct

Does the evaluation plan (federated vs. centralized baseline comparison) logically support the project's privacy-preservation claims? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☒ ☐ ☐ 4 - Completely correct

Are the limitations acknowledged by the team consistent with the architecture chosen (e.g., communication cost, non-IID challenges)? ***

0 1 2 3 4

0 - No valid ☐ ☐ ☒ ☐ ☐ 4 - Completely correct

Section 2: Logical Consistency

EXPERT CONSULTATION EVIDENCE (Cont...)

3. Validity and Justification

These questions probe whether design and evaluation decisions are adequately justified with evidence or precedent from the literature.

Is there sufficient justification for choosing BanglishBERT over alternatives such as mBERT or XLM-RoBERTa? *

0

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0 - No valid

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4 - Completely correct

Is the privacy-preservation claim adequately justified given that only architectural separation (no formal DP/encryption yet) is used? *

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0 - No valid

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4 - Completely correct

Is the validity of the simulated client partitioning method adequately explained and justified? *

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0 - No valid

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4 - Completely correct

Are performance metrics (accuracy, perplexity) appropriate and justified for evaluating next-word prediction specifically? *

0

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0 - No valid

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4 - Completely correct

4. Innovation and Contribution

These questions assess the originality and practical value of FedMed-BN relative to existing federated NLP and Bangla medical NLP research.

Does the project address a genuine gap in the literature, given that most federated NLP/medical studies focus on English (MIMIC-III, i2b2) or generic text classification benchmarks? *

0

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0 - No valid

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4 - Completely correct

Does the project respond meaningfully to future-work directions explicitly raised in prior literature (e.g., extending FL to low-resource/non-English languages)? *

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0 - No valid

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4 - Completely correct

Is there a clear, realistic practical application for this work in the Bangladeshi healthcare context? *

0

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0 - No valid

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4 - Completely correct

5. Overall Quality

These questions ask for a holistic judgment of technical soundness, coherence, and contribution of the thesis as a whole.

Overall, does the combination of FedAvg and BanglishBERT represent a technically sound and coherent solution to the stated problem? *

0

1

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0 - No valid

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4 - Completely correct

Are the project's stated limitations and future work realistic and proportionate to its current stage? *

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0 - No valid

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4 - Completely correct

Would this work be considered a meaningful and publishable contribution to federated NLP for low-resource medical text? *

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0 - Strongly Disagree

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4 - Strongly Agree

Comments or Additional Suggestions *

No comments

Section 3: Validity and Justification

Section 4: Innovation and Contribution

Section 5: Overall Quality